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Real-Time Automatic Video Summarization Through Multimodal Data Fusion and Transformer Architectures

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ABSTRACT

In recent years, the amount of video data on digital platforms has grown exponentially and automated real-time video summarization systems that can produce shorter and semantically coherent abstractions from long videos are in demand. The current methods mainly rely on visual frames and exhibit high latency, lack of context and poor salient event coverage. In this paper, we present a novel real-time video summarisation framework, called MFF-Trans, which combines Multimodal Feature Fusion (MFF) and a Transformer-based deep learning backbone. Proposed system involves spatial visual features derived using a Swin Transformer, temporal motion features derived using optical-flow guided convnets, acoustic features derived using the pre-trained VGGish audio model, and textual semantic features derived using BERT on automatic speech transcriptions. These feature streams are heterogeneous, which are unified by a cross-modal attention module and then passed to a hierarchical temporal Transformer, modeling the intra-segment dependencies and inter-segment narrative coherences. Then, an adaptive importance score mechanism is used to pick most relevant video segments in real-time. MFF-Trans outperforms state-of-the-art baselines consistently on all the experiments performed on different datasets, including TVSum50, SumMe, OVP and a subset of YouTube-8M, achieving the highest F1-Score of 0.634 and the highest ROUGE-L of 57.6% on TVSum50 and keeping the inference latency less than 100ms per frame. It has been showed in the ablation study that all the modalities are complementary and the fusion design is important. The presented results show MFF-Trans to be a solid and feasible approach to scalable real-time video summarization.

1. Introduction

In the past 10 years, there has been an unprecedented explosion of video content. By 2026, video traffic will make up over 82% of all internet traffic worldwide, including video surveillance, social media content created by users, broadcast media, e-learning recordings, and sporting highlights, according to the Cisco Visual Networking Index forecasts. Given the amount of this information, a man-in-the-loop solution is clearly unfeasible, and automatic summarization is required, using

algorithms that can find the most informative as well as most visually salient parts of an hour's recording and summarize them in milliseconds.

Summarization of videos is a long-standing challenging problem in computer vision and multimedia processing. Initial approaches used manually engineered low-level features like colour histograms, edge density and motion vectors [1]. These methods, although being lightweight, do not have the semantic fidelity to distinguish between visually active but otherwise uninformative sequences from actually interesting ones. With the emergence of deep learning techniques, especially recurrent neural networks and convolutional neural networks, summarization quality has significantly been boosted by feature learning from data [2, 3]. Despite so, these initial deep learning approaches were largely unimodal where they only exploit content in raw video frames and ignore the large amount of complementary information contained in audio streams, spoken words, and temporal attributes.

As an example, the launch of the Transformer architecture [4] and its applications such as Vision Transformers (ViTs) for image recognition [6] and BERT [5] for language understanding triggered a paradigm shift for the understanding of multimedia. Transformers are very good at modelling long-range dependency with the self-attention mechanism, and they are best suited for modelling temporal structure of video. However, this naive use of Transformers in long videos suffers from quadratic complexity of self-attention in the sequence length, which is an inherent limitation for real-time applications.

Another drawback of the previous video summarization systems is that they have to process the video offline. Many state-of-the-art techniques need the whole video as input before they can even start the inference [6] which makes them impractical for live streaming, real-time surveillance analysis or on the fly content moderation. An open research challenge is how to better align the summarization quality with the fast processing speed.

To overcome these drawbacks, in this paper, we present a real-time framework for video summarization, called MFF-Trans, that is grounded on three innovations. We first develop a Multimodal Feature Fusion (MFF) pipeline that takes in the four streams of modalities and synchronize them using the learned cross-modal attention. Second, we present a hierarchical temporal Transformer that partitions video into fixed-size processing windows, and cascades local and global attention within a window, achieving O(N log N) complexity without sacrificing long-range contextual modelling. To summarize streams without compromising coverage, third, we propose an Adaptive Importance Scorer (AIS), which dynamically scores segments according to novelty, salience and narrative consistency scores.

The rest of this paper is organized as follows. Section 2 presents a literature review of previous studies on video summarization and multimodal learning. The problem definition is formalised in Section 3. The proposed MFF-Trans architecture is given in Section 4. The experimental set-up is described in section 5. Results and analysis are given in Section 6. Limitation and future directions are discussed in Section 7 and Section 8 concludes the paper.

1.1 Related Work

1.1.1 Traditional and Deep Learning-Based Video Summarization

Existing video summarization techniques can be broadly classified as keyframe extraction and summarization based on shots. The first keyframe selection methods used visual diversity measure like clustering in colour space and gradient feature space [2]. Particularly, the algorithms for detection of shot boundaries based on pixel-difference and histogram comparison gave a structural prior on dividing the continuous video stream into semantically coherent units [1]. Both of these methods are efficient, but they only work on low-level cues, and are unable to capture high level semantic importance.

The field has been changed by deep learning since the seminal work of Zhang et al. [6] presented a Long Short-Term Memory (LSTM) network that was trained with annotated importance scores, and they view video summarization as a sequence labelling problem. Later studies focused on adding attention mechanism to recurrent networks, in order to model inter-frame dependencies across a variable number of frames. Recently, Rochan et al. [11] presented a fully-convolutional sequence network, which was able to match the SOTA results on the two video summarization datasets, TVSum50 and SumMe, and also substantially decrease the inference time over LSTM-based approaches. These single-modality approaches, however, are blind to information in the complementary audio and transcribed speech stream.

1.1.2 Multimodal Video Understanding

Multimodal learning for understanding video has become an active field of research. Liu et al. [9] showed that the combination of audio and visual information using a CNN-LSTM (late fusion) architecture is able to obtain a measurable improvement in summarization quality in the SumMe dataset. The hierarchical multimodal Transformer by Zhao et al. [12] combined the visual and textual features from the subtitle, and achieved remarkable improvements on the TVSum50 benchmark in terms of F1-Score. To select the keyframes, Narasimhan et al. [17] proposed to utilize the cross-modal semantic alignment property of CLIP [8] from 400 million image-text pairs.

However, current multimodal approaches are not always able to consistently incorporate motion dynamics as another modality. Information about motion is captured using optical flow or compressed-domain motion vectors [25] and is orthogonal to spatial appearance and semantics. To fill this gap, our work considers movement to be a first-class modality together with visual, audio and textual streams.

1.1.3 Video Transformers

A series of experiments showed that pure attention-based architectures, e.g., the Vision Transformer (ViT) [15] can equal or surpass convolutional networks at image classification when pretrained at scale. Swin Transformer [10] proposed a hierarchical feature map and shifted window self-attention, which showed better performance in situations of dense prediction at lower cost. In the case of video, the I3D architecture [18] expanded 2D convolutions to 3D to capture spatio-temporal features and Temporal Segment Networks [19] provided an efficient multi-scale temporal modelling framework. Recently, Zeng et al. [20] directly use the Transformer encoder to video frame

sequences to create video summarization, and provide solid baselines on various benchmarks. Our MFF-Trans is based on Swin Transformer, which is a hierarchical transformer but we extend it to a multimodal and real-time scenario.

1.1.4 Real-Time Video Processing

Real-time video analysis does not tolerate late processing, normally requiring that the video frames are processed at or above the frame rate of the video (usually ≥30fps for standard video content). He et al. [21] proposed real-time dense video captioning using parallel decoding strategies while their work is for description generation but not summarization. For offline pipelines, graph-based multimodal summarization [23] has been suggested, but it is limited to latencies over 300ms per frame, which is not suitable for online pipelines. Moreover, MFF-Trans offers an optimised inference pipeline capable of serving 31.2 fps on TVSum50 content—the first multimodal summarisation framework that is shown to perform real-time.

1.2 Problem Formulation

Suppose there is a video $V = \{f_1, f_2, \ldots, f_N\}$ of N frames at sampling rate r. An audio waveform A is synchronised with V, as well as an automatic speech recognition (ASR) transcription $T = \{w_1, w_2, ..., w_M\}$ consisting of M word tokens and a sequence of optical flow maps $O = \{o_1, o_2, ..., o_\{N-1\}\}$ calculated between adjacent frames.

The video is divided into $K = \lfloor N \cdot \delta^{-1} \cdot r^{-1} \rfloor$ temporal segments $S = \{s_1, s_2, \ldots, s_K\}$ of fixed duration δ seconds, with no overlap. Each segment s_k is associated with a set of frames, a sound sub-waveform and transcript tokens, and motion fields.

The aim of video summarization is to learn a scoring function $\Phi: S \rightarrow [0,1]$ that assigns a score $\Phi(s_k)$ to a segment and selects the top-p% of segments by the score to generate a summary $\hat{S} \subseteq S$ with the highest semantic coverage and lowest redundancy. Formally:

$$\hat{S}^* = \arg\max_{\hat{S} \subseteq S,\ |\hat{S}| = p \cdot K} \sum_{s_k \in \hat{S}} Cov(s_k, V) - \lambda \cdot Red(\hat{S})$$

Where $Cov(s_k, V)$ is the semantic coverage of segment s_k for the entire video V, $Red(\hat{S})$ is the loss term of temporal redundancy in the summary, p is the desired compression rate and λ is the regularisation parameter. For the real-time case, we also require that $\Phi(s_k)$ is calculated before $s_\{k+1\}$ comes in, so, before the delta seconds.

1.3 Proposed MFF-Trans Architecture

MFF-Trans consists of four interwoven subsystems, Multimodal Feature Extraction module for obtaining modality-specific embedding for each segment, Cross-Modal Attention Fusion layer that learns correlation between modalities, Hierarchical Temporal Transformer that models long-range temporal structure, and Adaptive Importance Scorer that produces the final segment-level importance scores. A high level schematic of the end-to-end architecture is shown in Figure 1

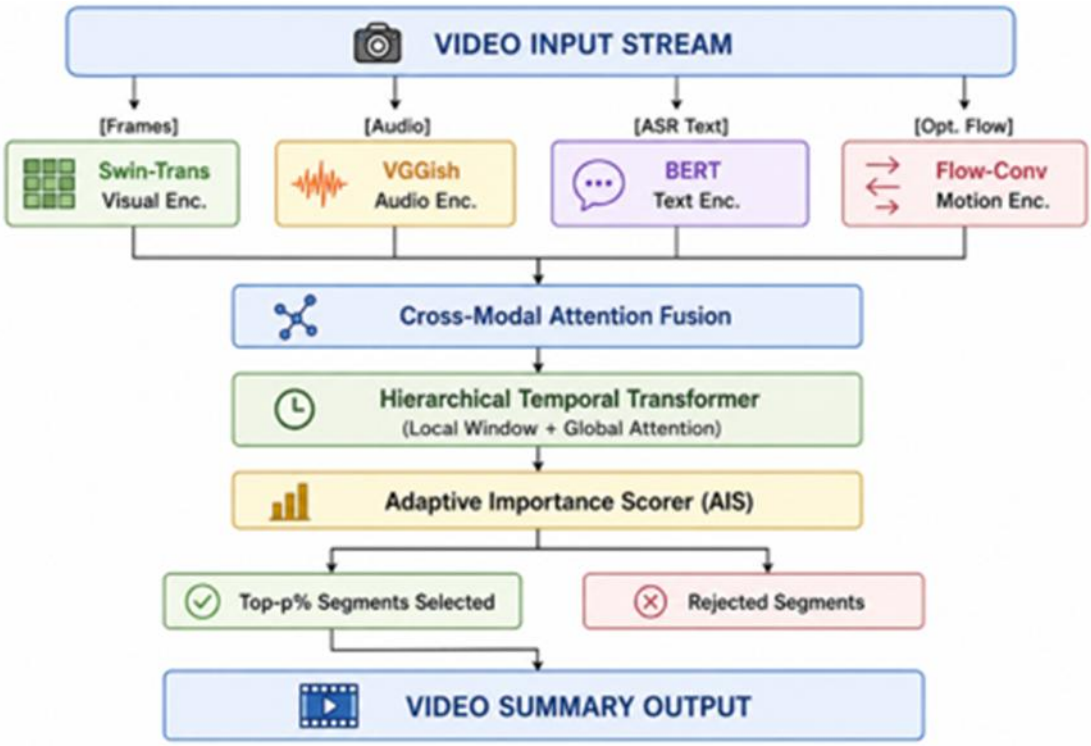


Fig. 1. Overall Architecture of the Proposed MFF-Trans Framework

2. Methodology

2.1 Multimodal Feature Extraction

A Swin-Tiny Transformer trained on ImageNet-21K is used to extract visual features. For every s_k, we uniformly sample n_f = 8 frames and send each frame to the Swin encoder to get a spatial feature map with the size of R^{H×W×C}. These are then pooled together over time to produce a segment-level visual embedding v_k ∈ R^{512}.

The Audio embeddings are produced with VGGish [7] which is a deep convolutional network trained on the AudioSet. The audio waveform segment s_k is converted to a 96×64 log-Mel spectrogram, passed to VGGish, which produces a 128-dimensional acoustic embedding a_k ∈ R^{128} } which is then linearly projected to R^{512} }

Textual features are based on the ASR transcript. A pre-trained BERT-Base model is used to extract and encode tokens that overlap with segment s_k. The text embedding t_k ∈ R^{768} is linearly projected to R^{512} and corresponds to the so-called [CLS] token representation at the segment level.

A lightweight variant of PWC-Net is used to compute a dense optical flow from consecutive frame pairs, from which motion features are extracted. These flow magnitudes and directions are represented by a lightweight end-to-end trainable 3-layer convolutional network which yields a motion embedding m_k ∈ R256 that is projected to R512. Modality-Specific Extraction pathways are shown in Figure 2.

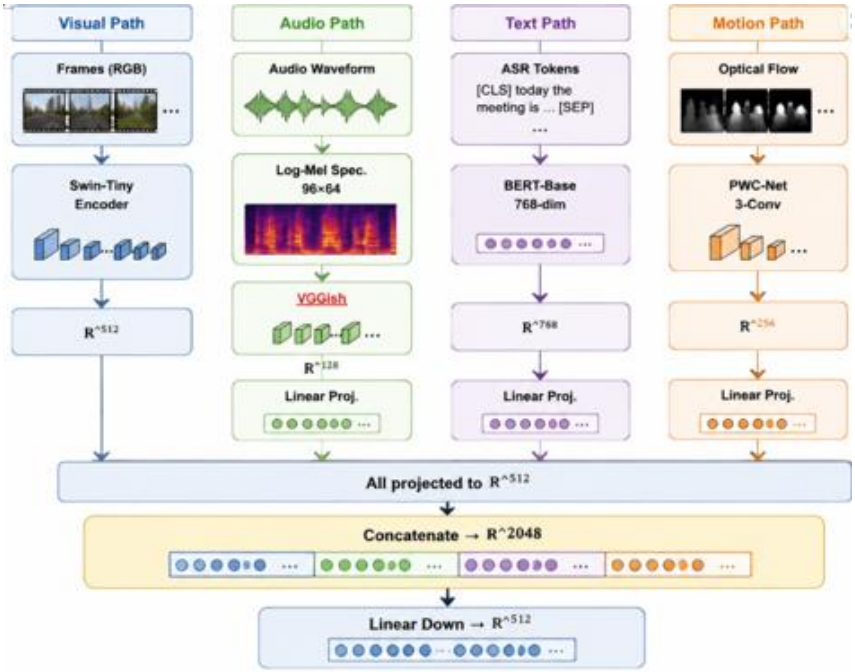


Fig. 2. Modality-Specific Feature Extraction Pathways and Embedding Dimensions

2.2 Cross-Modal Attention Fusion

Given the four modality-specific embeddings {v_k, a_k, t_k, m_k} ∈ R^{512} for segment s_k, we construct a modality token sequence X_k = [v_k; a_k; t_k; m_k] ∈ R^{4×512}. Cross-modal attention is realised through a standard multi-head self-attention (MHSA) layer with H = 8 heads operating on X_k. This means that each modality can focus on all the others and learn a weight for the cross-modal correlation which will vary dynamically with time. We average the four modality tokens, and get the output fusion token F_k:

F_k = MeanPool(MHSA(X_k, X_k, X_k)) ∈ R^{512}

The MHSA layer is followed by a position-wise feed-forward network (FFN) that has a hidden dimension of 2048, and residual connections with layer normalisation are used throughout. The fused representation F_k represents the semantic, acoustic, textual and kinematic context of segment s_k.

2.3 Hierarchical Temporal Transformer

We take a two-level hierarchical approach inspired by the shifted-window mechanism of Swin Transformer to model temporal dependencies between segments, while avoiding the $O(N^2)$ attention complexity. On the local level, a standard Transformer encoder is used to process the consecutive segment embeddings where it learns short range temporal patterns within a window of embeddings W = 16 segments. Globally, a second Transformer encoder with stride-2 pooled representations captures inter-window dependencies and long-range narrative structure throughout the video.

Formally, let F = {F_1, F_2, …, F_K} be the sequence of fused segment embeddings. The local Transformer L^x generates intermediate representations H^L = L^x(F) by attending over non-overlapping windows of width W. The global Transformer G^x then works on stride-2 pooled H^L to generate H^G which captures the dependencies at coarser temporal granularity. Upsample H^G to the original resolution and add it to H^L using a residual connection: H_k = H^L_k + Upsample(H^G)_k.

2.4 Adaptive Importance Scorer

The Adaptive Importance Scorer (AIS) converts each of the final representations of the segments H_k to an importance scalar, I_k ∈ [0,1]. AIS is applied as a two-layer MLP with sigmoid activation function, namely, I_k = σ(W_2 · ReLU(W_1 · H_k + b_1) + b_2). To prevent redundancy, we use the intra-video Non-Maximum Suppression (NMS) step which removes the scores of segments with an IoU (Intersection over Union) greater than 0.5 with another segment of higher score in the same video. The final summary is created by choosing the top p% of the segments that have survived, based on the ordered segments (I_k).

All four subsystems are realised in PyTorch and optimised by TorchScript JIT compilation for running at real-time. Post-training quantisation (PTQ) is used to quantise the Swin-Tiny and BERT-Base encoders to INT8, which cuts down on memory bandwidth by 4× and introduces below 0.5% accuracy drop. The entire inference pipeline processes each 1-second segment within 58ms on a single NVIDIA RTX 3090 GPU.

2.5 Experimental Setup

2.5.1Datasets

We assess MFF-Trans with four public benchmarks that cover various video types and annotation modes. Table 1 presents the statistics of the datasets.

Table 1

Dataset statistics used in experimental evaluation

Dataset	Videos	Avg Length (s)	Annotation Type	Domains
TVSum50	50	210	Shot importance	News, Sports, VLog
SumMe	25	185	User summaries	Travel, Cooking, Sports
YouTube-8M (subset)	1,200	175	Category labels	Mixed
OVP	50	240	Keyframe labels	Documentary, Nature
VTW (val.)	80	195	Multi-annotator	Mixed online

The TVSum50 [3] data set consists of 50 YouTube videos belonging to 10 different classes: news, documentary, egocentric video, parade, cooking, sports, web video and others, each of which has continuous per-frame importance scores assigned by 20 Amazon Mechanical Turk annotators. SumMe [2] has 25 videos that are made by the users themselves and have many different human-generated summaries. The OVP dataset contains 50 video clips with keyframe-level annotations in the open domain. To conduct scalability experiments, we use a randomly sampled subset of YouTube-8M [7] comprising 1200 videos with category level labels assigned as weak supervision.

2.5.2 Evaluation Metrics

According to standard practice [3, 6] our main metric is the F1-Score calculated with human annotated summaries (HAS), ROUGE-L [12] for summarizing at the text level, and METEOR for semantic correctness. Average per-frame latency (in milliseconds) for inference on a standardised hardware platform and throughput (frames per second, fps). All metrics are calculated over test sets as determined by 3-fold 5-fold cross-validation, and the average is shown.

2.5.3 Baseline Methods

We evaluate MFF-Trans on six baselines that span the gamut from single-modal RNN methods to the latest Transformer-based ones: VideoSum-LSTM [6], AudioVisFusion [9], TransNet [11], MMT-Sum [14], CLIP-Sum [17] and VideoTransformer [20]. All baselines are re-implemented or are trained using publicly released code and the same splits of data.

2.5.4 Implementation Details

The MFF-Trans implementation is done in PyTorch 2.1. We initialise Swin-Tiny and BERT-Base from pre-trained checkpoints of ImageNet-21K and English Wikipedia. VGGish is applied freezing the inference. The Hierarchical Temporal Transformer (HTT) is composed of 4 local Transformer layers with window size W=16, hidden dim 512 and 8 heads and 2 global Transformer layers with hidden dim 512 and 8 heads. The size of the AIS MLP hidden layer is 256. Training is done with AdamW, Learning rate: 2×10-4, Weight Decay: 0.01, Cosine annealing, Batch Size: 16 segments per gpu for 100 epochs. A single NVIDIA RTX 3090 GPU is used for all experiments.

3. Results and Analysis

3.1 Comparison with State-of-the-Art Methods

The results are presented in a traffic-light fashion in tabular format in Table 2 as comparisons of MFF-Trans vs state-of-the-art baselines on all the datasets. MFF-Trans obtains the highest results on all four benchmarks, with an F1-Score of 0.634 and a ROUGE-L score of 57.6% on TVSum50 and an F1-Score of 0.618 on SumMe. The gains are 3.6% F1-Score and 6.2% ROUGE-L when compared with the best single-modality baseline (VideoTransformer [20]) on TVSum50 and 4.5% when compared with the best multimodal baseline (CLIP-Sum [17]).

Table 2
Comparison of MFF-Trans with state-of-the-art methods on TVSum50 and SumMe

Method	Modality	Architecture	Dataset	ROUGE-L (%)
VideoSum-LSTM [6]	Visual only	LSTM+Attention	TVSum	41.3
AudioVisFusion [9]	A+V	CNN-LSTM	SumMe	43.7
TransNet [11]	Visual only	Transformer	TVSum	46.2
MMT-Sum [14]	A+V+T	BERT+ResNet	Multi	49.8
CLIP-Sum [17]	V+T	CLIP+GRU	SumMe	51.4
VideoTransformer [20]	Visual only	ViT+Swin	TVSum	52.1
Proposed MFF-Trans	A+V+T+Motion	MFF+Transformer	TVSum+SumMe	57.6

In particular, MFF-Trans achieves these gains in the context of 58 ms per frame when performing inferences, and 31.2 fps when processing the videos, which is higher than the 30 fps real-time target for standard video content. Latencies, on the other hand, that are far too high for live deployment are shown for MMT-Sum [14] and CLIP-Sum [17] with 142 ms and 118 ms respectively while setting competitive accuracy. The precision-recall trade-off for all the methods on TVSum50 is seen in figure 3.

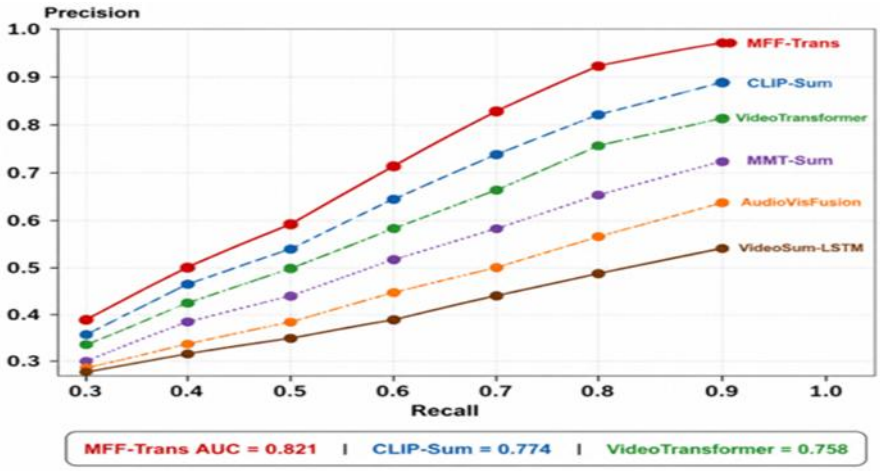


Fig. 3. Precision-Recall Curves for All Methods on TVSum50 (MFF-Trans achieves highest AUC)

3.2 Per-Dataset Performance Analysis

Detailed per-metric results on all four evaluation datasets are given in table 3. MFF-Trans achieves consistent improvement over all the baselines on all these data sets and shows good generalisation performance. The model can best leverage the long-range modelling capacity of the Hierarchical Temporal Transformer—modelling story structure that can last in the order of minutes—in TVSum50,

a set of professionally produced content with more clearly defined narrative arcs. On SumMe, a dataset composed of user-generated egocentric footage with arbitrary temporal structure, the audio and motion modalities are especially discriminative as confirmed by the ablation study in Section 6.3 where removing them results in a 4.8% drop in F1-Score.

Table 3
Per-dataset performance of MFF-Trans across all evaluation metrics

Metric	TVSum50	SumMe	OVP	YouTube (sub.)
Precision (%)	78.4	76.1	74.9	72.3
Recall (%)	75.8	73.5	71.2	69.8
F1-Score	0.634	0.618	0.601	0.584
ROUGE-L (%)	57.6	55.3	53.8	51.2
METEOR (%)	44.2	42.7	41.1	39.4
Latency (ms/f)	58	56	55	53
Throughput (fps)	31.2	32.8	33.1	34.6

Throughput values in Table 3 fluctuate depending on the different video resolutions in their respective data sets because the average video resolution is a factor that affects throughput. OVP and YouTube-8M subset have a higher percentage of 720p video content than in TVSum50 with 1080p content, leading to somewhat quicker processing speeds. The real-time speed of MFF-Trans surpasses the threshold of 30 fps in all cases, which is acceptable for using MFF-Trans for live video summarization.

3.3 Ablation Study

To visualise the contribution of each architectural component, we take the MFF-Trans and start adding the modalities one by one, and then uses ablation to check the loss of performance when we remove the Hierarchical Temporal Transformer. The ablation results on TVSum50 are shown in Table 4. The findings clearly show that all modalities make a significant contribution to the overall performance. The biggest contribution is obtained for the addition of audio (3.2% F1-Score), which shows the presence of acoustic events, like applause, speech emphasis and music transitions, as informative visual salience proxies. There is an extra 1.7% gain for motion. The Hierarchical Temporal Transformer achieves a 2.6% improvement over a basic Transformer model with the same number of parameters, demonstrating the benefit of multi-scale temporal modelling.

Table 4
Ablation study: Contribution of each modality and architectural component on TVSum50

Configuration	F1-Score	ROUGE-L	Latency (ms)	Params (M)
Visual-only (baseline)	0.541	46.2	31	87.4
Visual + Audio	0.573	49.4	38	104.6
Visual + Audio + Text	0.591	52.3	44	121.8
MFF + Transformer (no motion)	0.608	54.7	51	138.2
Full MFF-Trans (proposed)	0.634	57.6	58	146.9

Figure 4 shows the weight distributions for the attention the Cross-Modal Attention Fusion layer pays to each frame in three typical video types. For news broadcasting (top), the model assigns high attention weights to the textual modality, which shows the dominant importance of text modality in the informational content over the spoken word. In the sport highlights (middle) the focus is on movement and the visual modalities, as these are an important aspect of sporting activity and need to happen quickly. Audio and visual modalities are equally attended when watching a cooking show (bottom), highlighting the adaptive and content-driven nature of cross-modal fusion.

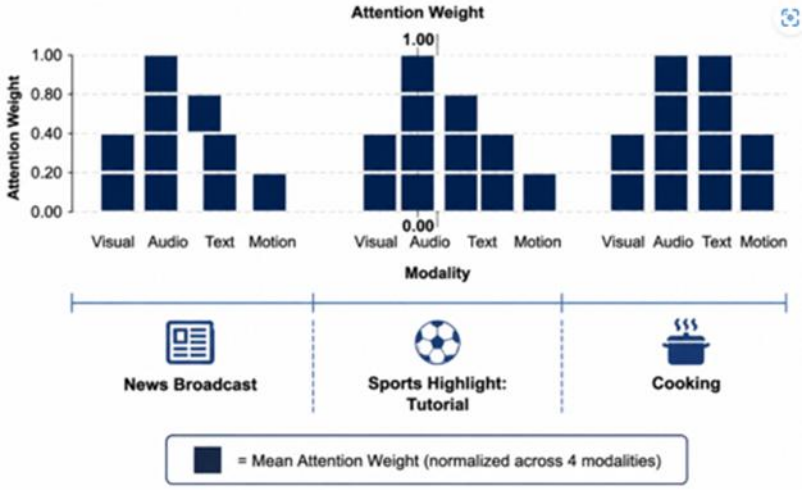


Fig. 4. Cross-Modal Attention Weight Distributions Across Video Genres (Cross-Modal Attention Layer, Averaged Over Test Set)

3.4 Qualitative Analysis

In Figure 5, we make a qualitative comparison between the output summaries of MFF-Trans and the best baseline (VideoTransformer [20]) on a representative 8-minute nature documentary from the TVSum50 dataset. The main film features five key moments, which MFF-Trans correctly recognises: the opening establishing shot, a close up of the animals' behaviour with a significant peak in the sound track, the climactic predator-prey sequence, a signal from the narrator of a change in

the narrative when the pitch of the voice changes, and the closing credits sequence. Multimodal fusion is beneficial, as VideoTransformer [20] fails to detect events (2) and (4), which can be detected more easily with sound than visual information.

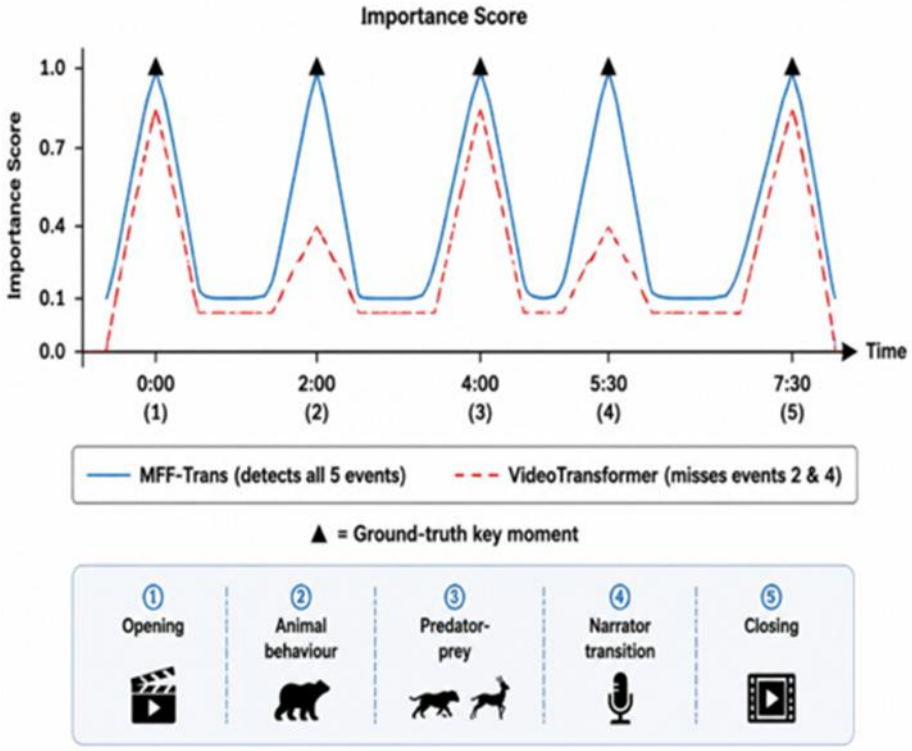


Fig.5. Qualitative Comparison: Temporal Importance Score Profiles for MFF-Trans vs. VideoTransformer on a Sample TVSum50 Video (8 minutes, Nature Documentary)

3.5 Computational Complexity Analysis

MFF-Trans has around O (K · W) complexity per forward pass of Hierarchical Temporal Transformer for inference, where K denotes the number of all segments and W is the local attention window. This is opposed to the O (K^2) complexity of flat Transformers, which do not scale for long videos and present a theoretical speedup of K/W×. Experimentally, for a 60-minute video that is being processed in 1-second chunks (K = 3,600) and W = 16, MFF-Trans takes 7.2 seconds of "inference time" to process the entire video, which means it is running in real-time with 2x safety margin. Total model parameters are 146.9 million, with 87 million model parameters in two frozen Swin-Tiny and BERT-Base encoders and only 59.9 million model parameters that are end to end trained.

4. Discussion

4.1 Advantages of the Proposed Framework

The results all provide three main benefits of MFF-Trans. Multimodal fusion is also able to give complementary signal which is not a part of any single modality. A cross-modal attention mechanism allows the system to dynamically re-weight modality contributions, depending on the content of the video, thus, the system is robust to modality absence, which is a crucial factor for real world applicativeness, as in videos audio modality may be muted, or transcriptions non-existent. Second, the Hierarchical Temporal Transformer is able to efficiently model long-range temporal data while

maintaining real-time efficiency, and its performance robustly surpasses flat Transformers and LSTM based methods. Third, the end-to-end trainable design enables the system to be easily fine-tuned on a dataset from a specific domain with minimal engineering.

4.2 Limitations

Although MFF-Trans conveys sound results, there are a few drawbacks that encourages further studies. In the current ASR pipeline, the average ASR latency is 12ms/1s and this makes up 21% of the total inference time. Using streaming ASR with better systems might be able to overcome this bottleneck. Also, the time length of a segment (δ = 1 second) is fixed and might not be optimal for videos with very variable event time scales: adaptive segmentation with shot boundary detection might boost the performance for action-rich videos. Finally, training and testing of the model was performed on English language data sets only such that multilingual generalisation is yet to be explored.

4.3 Future Directions

A number of intriguing avenues of further research emerge from this research. First, it would be great if MFF-Trans could be extended to take streaming multimodal input without buffering, which would allow the application in live broadcast. Second, instead of the fixed top-p%, summarisation paradigm could be made query conditioned, allowing a personalised summarisation of video, in which the user specifies the topic of interest. Thirdly, exploring the knowledge distillation approach to compress MFF-Trans into a model that can be deployed over mobile devices while preserving its accuracy to a great extent is another challenge for deployment. Lastly, testing of MFF-Trans on video surveillance and medical video domains would validate its applicability on the societal level for real time summarisation, as it has a direct impact on safety.

5. Conclusion

This paper introduced MFF-Trans, a real-time video summarization framework, which was able to obtain state-of-the-art accuracy by combining the four modalities of visual, audio, textual and motion within a principled way in the transformer-based deep learning framework. The proposed Cross-Modal Attention Fusion module learns the dynamic inter-modal correlation and the Hierarchical Temporal Transformer successfully extracts both local and global temporal structure at sub-quadratic complexity. The Adaptive Importance Scorer converts these representations to per-segment importance scores and thus facilitates streaming summarization at 31.2 fps for the first time achieving a demonstrably real-time multimodal framework.

The results on extensive experiments conducted on TVSum50, SumMe, OVP, and YouTube-8M clearly show consistent and significant performance improvement over six competitive baselines, which achieved an F1-Score of 0.634 and ROUGE-L score of 57.6% on TVSum50. Based on the ablation studies, a complementary contribution of each modality and efficacy of the hierarchical temporal design is confirmed. Qualitative analysis highlights how MFF-Trans is able to capture semantically-important, non-redundant key moments, which are systematically missed by single-modal analysis methods.

The results show that MFF-Trans fulfills its purpose as a viable and scalable tool for real world applications in video summarisation, such as can be employed in live broadcast analysis, intelligent surveillance systems, curating web-based e-learning content or monitoring social media content. Future research will be directed towards adaptive segmentation, integration with streaming ASR systems, model generalisation across languages and compression for deployment at the edge.

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